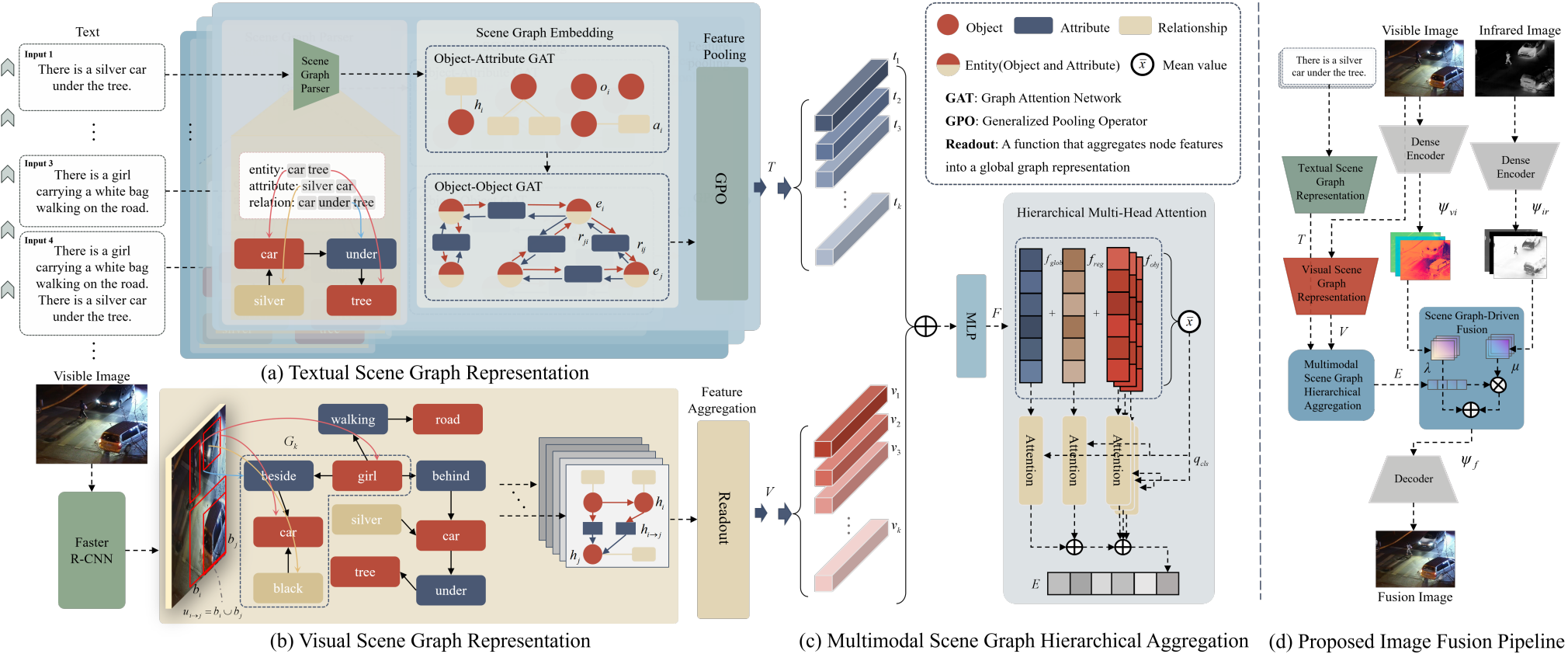
Text
Input 1
There is a silver car under the tree.
Input 3
There is a girl carrying a white bag walking on the road.
Input 4
There is a girl carrying a white bag walking on the road. There is a silver car under the tree.
Scene Graph Parser
entity: car tree
attribute: silver car
relation: car under tree
car
under
silver
tree
Scene Graph Embedding
Object-Attribute GAT
Object-Object GAT
Feature Pooling
GPO
T
(a) Textual Scene Graph Representation
Visible Image
Faster R-CNN
walking
road
beside
girl
behind
car
silver
car
black
tree
under
Feature Aggregation
Readout
V
(b) Visual Scene Graph Representation
Object
Attribute
Relationship
Entity(Object and Attribute)
Mean value
GAT: Graph Attention Network
GPO: Generalized Pooling Operator
Readout: A function that aggregates node features into a global graph representation
MLP
F
Hierarchical Multi-Head Attention
Attention
Attention
Attention
E
(c) Multimodal Scene Graph Hierarchical Aggregation
Visible Image
Infrared Image
There is a silver car under the tree.
Textual Scene Graph Representation
Dense Encoder
Dense Encoder
Visual Scene Graph Representation
Scene Graph-Driven Fusion
Multimodal Scene Graph Hierarchical Aggregation
Decoder
Fusion Image
(d) Proposed Image Fusion Pipeline